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Cases: For liquids:

$$v = \sqrt{\frac{K}{\rho}} \qquad (11.19)$$

where, K is the bulk modulus and ρ is the density of the rod.

EXAMPLE 11.7

Calculate the speed of sound in a steel rod whose Young's modulus $Y = 2 \times 10^{11}$ N m^{-2} and $\rho = 7800\ kg\ m^{-3}$.

Solution

$$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2\times10^{11}}{7800}} = \sqrt{0.2564\times10^{8}}$$

$$= 0.506\times10^{4}\, m s^{-1} = 5\times10^{3}\, m s^{-1}$$

Therefore, longitudinal waves travel faster in a solid than in a liquid or a gas. Now you may understand why a shepherd checks before crossing railway track by keeping his ears on the rails to safegaurd his cattle.

EXAMPLE 11.8

An increase in pressure of 100 *k*Pa causes a certain volume of water to decrease by 0.005% of its original volume.

(a) Calculate the bulk modulus of water?.

(b) Compute the speed of sound (compressional waves) in water?.

Solution

(a) Bulk modulus

$$B = V\left|\frac{\Delta P}{\Delta V}\right| = \frac{100\times10^{3}}{0.005\times10^{-2}} =$$

$$= \frac{100\times10^{3}}{5\times10^{-5}} = 2000\, MPa,$$ where MP_a mega pascal

(b) Speed of sound in water is

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{2000\times10^{6}}{1000}} = 1414\, m s^{-1}$$

Note

The velocities of both transverse waves and longitudinal waves depend on elastic property (like string tension T or bulk modulus B) and inertial property (like density or mass per unit length) i.e., $v = \sqrt{\frac{Elastic\ property}{Inertial\ property}}$

Table 11.2: Speed of sound in various media

S.No.	Medium	Speed in m s^{-1}
	Solids	
1.	Rubber	1600
2.	Gold	3240
3.	Brass	4700
4.	Copper	5010
5.	Iron	5950
6.	Aluminum	6420
	Liquids at 25°C	
1.	Kerosene	1324
2.	Mercury	1450
3.	Water	1493
4.	Sea Water	1533
	Gas (at 0°C)	
1.	Oxygen	317
2.	Air	331
3.	Helium	972
4.	Hydrogen	1286
	Gas (at 20°C)	
1.	Air	343